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Lyapunov-like conditions for prescribed-time stability of perturbed impulsive systems

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ABSTRACT

The present work deals with the problem of prescribed-time control of non-linear impulsive systems consisting of external perturbations. Lyapunov-like sufficient conditions for prescribed-time and practical prescribed-time stability are provided for vanishing and non-vanishing perturbations, respectively. Depending on the user's requirements, some sequences of stabilizing impulses are constructed in this regard. It is shown that the systems consisting of vanishing-type disturbances can attain prescribed-time stabilization at the origin. On the other hand, in the presence of non-vanishing disturbances, which can consist of bounded or **unbounded** disturbances, the trajectories of the system can enter a stable region only within the prescribed time. Moreover, the stable region is independent of the impulsive strength and the prescribed-time. The efficacy of the proposed results is provided through various examples and their numerical simulations.

1. Introduction

Impulsive systems, an essential class of hybrid systems, have drawn substantial attention from researchers due to their wide range of applications in the fields of engineering, biology, physics, and control technology. The pioneering work in the field of impulsive systems, which works as a bridge between continuous and discrete systems, can be found in the book [1]. T. Yang established the impulsive control theory in his book [2]. There are two types of stability problems associated with impulsive systems: impulsive control problems and impulsive perturbation problems. Stabilization of a system through stabilizing impulses is named as impulsive control problem. An impulse perturbation problem, on the other hand, involves studying the robustness of a system containing destabilizing impulses. Impulsive control problems have grown leaps and bounds recently because of their discontinuous nature, which helps to reduce the control cost and increases robustness. Several works in this regard can be found in pest management [3], orbit transfer [4], hyperchaos synchronization [5], and secure communication [6]. Further, various advancements in the theory of impulsive control can be noted in some recent articles (see [7–10] and references therein). Specifically, [7] studied the stability properties of non-linear systems by considering state-dependent delay into account. Meanwhile, [10] provides stability properties of infinite dynamical systems that can deal with continuous and discrete dynamics instability, concurrently.

It is worth recalling that the notion of finite-time stability (*FTS*) has captured the attention of many researchers because of its noticeable properties, such as the capability of rapid convergence to the desired region and robustness against disturbances. The revolutionary work in this regard can be found in [11–13]. In the following years, several other such results on *FTS* for numerous models have been presented in [14–17]. In particular, [14] proposed some results and discussed their applications for

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FTS of time-delayed systems using Lyapunov functionals. Whereas, some Lyapunov-based sufficient conditions for impulsive systems are provided in [17], considering both stabilizing and destabilizing impulses. They also estimated the settling time, which depends not only on the initial conditions but also on the impulses. In spite of several advantages of FTS, it suffers from the drawback of dependency of the settling-time on the initial conditions. To overcome the shortcomings of FTS, Polyakov proposed a new notion of stability called fixed-time stability (F_XTS) in his article [18]. Here the upper-bound of the settling-time depends on the systems' parameters only and is independent of the initial conditions. A number of works on this aspect are available in [19–21], along with the references therein. Specifically, [19] has provided a global robust fixed-time stability result using variable exponent coefficients and then the result is applied to stabilize a second order uncertain system through sliding mode control. Whereas, [20] proposed some sufficient conditions for F_XTS of impulsive systems considering both stabilizing and destabilizing impulses. Meanwhile, there are many practical applications in which it is necessary to predetermine the upper-bound of the settling-time in advance. This is not always feasible when it comes to F_XTS . Thus, a new concept of stability, predefined-time stability (P_DTS) was introduced in the article [22]. Following that, [23–25] extended and modified this concept in several models. For example, [23] and [24] have provided unified Lyapunov-like sufficient conditions and showed their applications in sliding mode control. And, [25] has provided some sufficient conditions for F_XTS and P_DTS of impulsive systems considering destabilizing and hybrid impulses. It should be pointed out that P_DTS of nonlinear systems is not relevant if the system's parameters are inaccessible to be altered. Hence, Song and co introduced a novel form of stability, prescribed-time stability (PTS) in [26]. The main attraction of PTS is that the user can preset the upper-bound of the settling-time, which is independent of the system's parameters and initial conditions. The research community has been educated by the writers of [27–29] and the references included in them by means of their creative presentation of PTS and its use across different models. In particular, [27] provided a sufficient condition for PTS of a class of uncertain non-linear time-delay systems using the Lyapunov–Krasovskii functional method. A novel approach for PTS is given in [28], whereby, by means of a proper time scale deformation, an infinite interval is squeezed into a prescribed one, and several results have been discussed on the basis of this concept. Additionally, [29] proposed a local PTS condition corresponding to unstable dynamics based on impulsive control theory. Meanwhile, though the concept of PTS for impulsive systems or PTS by means of impulsive control is in its infancy, some of the contributing articles in this field can be found in [29–32]. In brief, [30] provided a new PTS theorem for nonlinear impulsive piecewise-smooth systems and is applied to Chua's circuit. Whereas some results concerning the PTS of switched-systems by means of event-triggered impulsive control are depicted in [31]. And, PTS of switching complex networks under the effects of impulses is studied in [32].

It is noteworthy to mention that none of the articles mentioned above have accounted for the external disturbances. However, in practical models, the influences of external behaviours are inevitable. For example, robots in robotics frequently encounter disruptions like unforeseen obstructions. Whereas, several disturbances like earthquakes or high winds in civil engineering may effect the structural integrity of buildings. And in transportation engineering, disturbances like traffic jams, severe weather, and accidents may affect the practical implementation of the model. In light of this, it is imperative from a practical standpoint to take disturbances into consideration and develop several approaches to eliminate them through well-posed control. Meanwhile, disturbances are categorized as vanishing and non-vanishing. In case of vanishing disturbances, the trajectories of the systems can reach the origin within finite/fixed/predefined or prescribed-time. However, the system states can only reach some neighbourhood of the origin in case of non-vanishing disturbances, called practical-finite/fixed/predefined or prescribed-time stability. [33–35] and the references therein are some useful articles in this regard. In brief, a modified Lyapunov form, $\dot{V}_p \leq -a_1 V_p^{b_1} + c_1$ with $a_1, c_1 > 0$ and $0 < b_1 < 1$ was studied in [33] for attitude control of a spacecraft model under the influence of external disturbances. Whereas, [35] used another modified Lyapunov form, $\dot{V}_p \leq -a_0 V_p - a_1 V_p^{b_1} + c_1$ with $a_0, a_1, c_1 > 0$ and $0 < b_1 < 1$, to study finite-time tracking control of quadrotor Unmanned Aerial Vehicle (UAV) system under the effects of modelling uncertainties and external disturbances. Recently, the concept of FTS properties of impulsive systems under the effect of disturbances has been presented in [36,37]. Precisely, the authors of [36] have provided some Lyapunov-based sufficient conditions for disturbed impulsive systems considering both vanishing and non-vanishing disturbances. In the case of vanishing disturbances, they provided some conditions for FTS, whereas practical finite-time stability ($PFTS$) was studied in the case of non-vanishing disturbances. Meanwhile, [37] studied $PFTS$ of non-linear systems under the effect of delayed impulsive control. It should be noted that both of these articles ([36,37]) have accounted for bounded non-vanishing disturbances. Afterwards, it is natural to wonder if similar outcomes are possible for unbounded non-vanishing disturbances. Further, initial conditions are involved in both of these two articles. Hence, it is a challenging work if one can propose some PTS results for disturbed non-linear impulsive systems. As far as we know, no article in the existing literature has considered these aspects.

Motivated by the above discussions, some Lyapunov-like conditions for both PTS and practical prescribed-time stability ($PPTS$) that correspond to non-linear impulsive systems are proposed in this article. The main contributions of the article can be outlined as follows:

1. With the help of the Lyapunov method, the stability problem of perturbed non-linear systems is studied via an impulsive control strategy.
2. Compared to the recent articles ([36,37]), which studied FTS and $PFTS$, we have studied PTS and $PPTS$ of non-linear impulsive systems, where the global Lipschitz condition of the Lyapunov function, provided in [36], has been relaxed in this article.
3. Some Lyapunov-like sufficient conditions for uniform global PTS and $PPTS$ are proposed for perturbed non-linear impulsive systems consisting of vanishing and non-vanishing disturbances, respectively.

4. In contrast to the existing articles, which contain only bounded non-vanishing disturbances, our results can also counter a class of *unbounded non-vanishing disturbances* and provide a uniform global *PPTS* of the impulsive system.

The rest of the article is organized as follows: Section 2 is dedicated to the introduction of some helpful notation, system descriptions, and definitions. Section 3 provides the article's primary results, consisting of uniform global PTS and PPTS theorems for perturbed non-linear impulsive systems. Section 4 illustrates the validity of the proposed results through four examples. Section 5 concludes the article by highlighting the significant contributions and scope for future work.

2. Preliminaries

A variety of notations, system's description and some definitions are included in this section, which serves as the preliminaries to the rest of the article.

2.1. Notations

- $\mathbb{R}$ is the set of real numbers; $\mathbb{R}_+ = \{y \in \mathbb{R} : y > 0\}$; $\mathbb{R}_{\geq 0} = \mathbb{R}_+ \cup \{0\}$.
- $\mathbb{R}^n$ is the n -dimensional real space with the Euclidean norm $|\cdot|$.
- $\mathbb{N}$ is the set of positive integers.
- $\text{sgn}(y)$ represents the signum function for $y \in \mathbb{R}$.
- ∇ is the gradient operator.

2.2. System's description

Let us consider the nonlinear disturbed impulsive system:

$$\begin{cases} \dot{y}(t) = f(y(t)) + \varpi(t, y(t)), t \geq 0, t \neq s_k, \\ y(t) = g(y(t^-)), t = s_k, \\ y(0) = y_0, \end{cases} \quad (1)$$

where $y \in \mathbb{R}^n$ is the state vector; $\dot{y}(t)$ is the right hand derivative of $y(t)$; f, g are the continuous functions from $\mathbb{R}^n$ to $\mathbb{R}^n$ satisfying $f(0) = g(0) = 0$ for all $t \geq 0$. Further, $\varpi : \mathbb{R}_{\geq 0} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes the disturbance present in the system and $\{s_k, k \in \mathbb{N}\}$ (in short $\{s_k\}$) is either finite or infinite impulsive sequences satisfying $0 < s_1 < s_2 < \dots < s_k \rightarrow +\infty$ as $k \rightarrow +\infty$. Let $\mathcal{H}$ denote such a class of impulsive sequences. It is also assumed that f and g fulfil all the conditions, that is, when $\varpi \equiv 0$, that ensures the unique existence of the solution $y(t) = y(t, y_0)$ of the system (1) in forward time (see [2]). In addition, without losing generality, it is assumed that the solution $y(t)$ is right continuous and has left limits at all times.

2.3. Some definitions

Definition 1. [29] For any $T_p > 0$, a prescribed constant, the origin of system (1) is said to be uniformly globally prescribed-time stable (U_GPTS) with respect to T_p over the class of impulsive sequence $\mathcal{H}_0 \subseteq \mathcal{H}$, if

- for any $\emptyset > 0$ and $\{s_k\} \in \mathcal{H}_0$, there exists $\mathcal{L}(\emptyset) > 0$ such that whenever $|y_0| < \mathcal{L}(\emptyset)$ holds, it implies $|y(t)| < \emptyset \forall t \geq 0$, and
- for any $y_0 \in \mathbb{R}^n$ and $\{s_k\} \in \mathcal{H}_0$, $y(t, y_0) = 0 \forall t \geq T_p$.

Definition 2. For given $\chi > 0$ and any prescribed constant $T_p > 0$, the origin of system (1) is said to be uniformly globally practical prescribed-time stable (U_GPPTS) with respect to T_p and χ over the class of impulsive sequence $\mathcal{H}_0 \subseteq \mathcal{H}$, if for any $y_0 \in \mathbb{R}^n$ and $\{s_k\} \in \mathcal{H}_0$, $|y(t, y_0)| \leq \chi \forall t \geq T_p$.

Remark 1. [29] The term “uniformly” in Definition 1 and 2 is used to mean that the prescribed-time property of system (1) is independent of the choice of the impulsive sequence $\{s_k\} \in \mathcal{H}_0$. Further, Definition 1 is different from several notions of stability like-fixed-time and predefined-time as the bound of the settling time can be preassigned arbitrarily and also does not depend on the parameters of the system, respectively.

Definition 3. [29] (κ_∞ Functions): A strictly increasing continuous function $\psi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be a κ_∞ function if $\psi(0) = 0$ and $\psi(a) \rightarrow \infty$ as $a \rightarrow \infty$. And is denoted by $\psi \in \kappa_\infty$.

3. Main results

In this section of the manuscript, based on the Lyapunov method and impulsive control theory, two theorems are provided for U_GPTS and U_GPPTS of the system (1) corresponding to vanishing and non-vanishing (*bounded and unbounded*) disturbances respectively.

Theorem 1. For any prescribed-constant $T_p > 0$, let there exist κ_∞ functions σ_1, σ_2 , constants $\alpha, A, B > 0, 0 < c, p, q < 1$ and a locally Lipschitz continuous function $V_P : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$, such that:

- i) $\sigma_1(|y|) \leq V_P(y) \leq \sigma_2(|y|)$;
 - ii) $\xi(V_P(g(y))) \leq c\xi(V_P(y))$ for $y \in \mathbb{R}^n$;
 - iii) $\frac{d\xi}{dV_P} \nabla V_P(y) \cdot f(y) \leq -A\xi^p(1-\xi)$ for almost all $y \in \mathbb{R}^n$;
 - iv) $\frac{d\xi}{dV_P} \nabla V_P(y) \cdot \varpi(t, y) \leq B\xi^p(1-\xi)^q$ for almost all $y \in \mathbb{R}^n$ and all $t \geq 0$;
- where, $\xi \equiv \xi(V_P) = \frac{\alpha V_P}{1+\alpha V_P}$.

Then, system (1) is U_G PTS with respect to prescribed-constant T_p over the class of impulsive sequences $\mathcal{H}_\diamond \subseteq \mathcal{H}$, where $\mathcal{H}_\diamond$ denotes the set of impulsive sequences $\{s_k\}$ with $N + N_0$ as the minimum number of impulses satisfying

$$s_N < \mu T_p, \quad (2)$$

$$N = \min \left\{ i \in \mathbb{N} \cup \{0\} \mid i \geq \frac{\ln \left(1 - (kl)^{\frac{1}{1-q}} \right) - B' \mu T_p}{\ln(c)} \right\}, \quad (3)$$

with $0 < \mu < 1$, $l = \frac{B}{A} < 1$, $k > 1$, $kl < 1$, $B' = \frac{B(kl)^{\frac{q}{1-q}}}{\left(1 - (kl)^{\frac{1}{1-q}} \right)^{1-p}}$ and

$$\mu T_p < s_{N+1} < \dots < s_{N+N_0} \leq T_p, \quad (4)$$

$$s_{N+i} - s_{N+i-1} \geq c^{(i-1)(1-p)}, i \in \{2, 3, \dots, N_0\}, \quad (5)$$

with

$$N_0 = \min \left\{ i \in \mathbb{N} \cup \{0\} \mid i \geq 1 + \frac{\left(1 - (kl)^{\frac{1}{1-q}} \right)^{1-p}}{Al(k-1)(1-p)(kl)^{\frac{q}{1-q}}} \right\}. \quad (6)$$

Proof. Let $T_p > 0$ be given. Then, for any initial value $y_0 \in \mathbb{R}^n$ and impulsive sequence $\{s_k\} \in \mathcal{H}_\diamond$, $y(t) = y(t, y_0)$ is considered as the solution of the system (1) through $(0, y_0)$.

The proof of the theorem is divided into two cases as, $\xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}$ and $\xi(t) > 1 - (kl)^{\frac{1}{1-q}}$, where, $\xi_0 \equiv \xi(V_P(y_0))$ and $\xi(t) \equiv \xi(V_P(y(t)))$.

Case 1: $\xi_0 \leq 1 - (kl)^{\frac{1}{1-q}}$:

For this case, the aim is to show that $\xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}$ for all $t \geq 0$. If this is false, then there is some $t > 0$ such that $\xi(t) > 1 - (kl)^{\frac{1}{1-q}}$.

Now let us denote $t^\circ = \inf \left\{ t \geq 0 : \xi(t) > 1 - (kl)^{\frac{1}{1-q}} \right\}$.

As $c \in (0, 1)$, $t = t^\circ$ can not be an impulsive instant. Hence one can have, $\xi(t^\circ) = 1 - (kl)^{\frac{1}{1-q}}$, $\xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}$ for all $t \in [0, t^\circ]$ and the derivative of ξ along the solution trajectories of the system (1) at $t = t^\circ$ satisfies $D^+\xi(t)|_{t=t^\circ} \geq 0$.

Moreover, depending on the conditions (iii) and (iv) one can have

$$\begin{aligned} D^+\xi(t)|_{t=t^\circ} &\leq -A\xi(t^\circ)^P(1-\xi(t^\circ)) + B\xi(t^\circ)^P(1-\xi(t^\circ))^q \\ &= A\xi(t^\circ)^P(1-\xi(t^\circ))^q \left\{ - (1-\xi(t^\circ))^{1-q} + \frac{B}{A} \right\} \\ &= A\xi(t^\circ)^P(1-\xi(t^\circ))^q \{-kl + l\} \\ &= -Al\xi(t^\circ)^P(1-\xi(t^\circ))^q(k-1) \\ &< 0. \end{aligned}$$

This is a contradiction. Hence, $\xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}$ for all $t \geq 0$ holds and thereupon

$$D^+\xi(t) \leq -Al\xi(t)^P(1-\xi(t))^q(k-1) \forall t \geq 0, \quad (7)$$

which implies the Lyapunov stability of the system (1). Moreover, using $\xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}$ for all $t \geq 0$, one can have from eqn. (7) that

$$\begin{aligned} D^+ \xi(t) &\leq -Al(kl)^{\frac{q}{1-q}}(k-1)\xi(t)^P, \forall t \geq 0 \\ &= -L'\xi(t)^P, \forall t \geq 0, \end{aligned} \quad (8)$$

where $L' = Al(kl)^{\frac{q}{1-q}}(k-1)$.

From eqn. (8) one can easily have

$$\xi^{1-p}(t) \leq \xi_0^{1-p} - L'(1-p)t, \forall t \in [0, s_1]. \quad (9)$$

Moreover, using condition (ii) one can have at $t = s_1$

$$\begin{aligned} \xi^{1-p}(s_1) &\leq c^{1-p} \left\{ \xi_0^{1-p} - L'(1-p)s_1 \right\} \\ &\leq \xi_0^{1-p} - L'(1-p)s_1. \end{aligned} \quad (10)$$

Similarly on the interval $[s_1, s_2)$ one can deduce

$$\begin{aligned} \xi^{1-p}(t) &\leq \xi^{1-p}(s_1) - L'(1-p)(t-s_1) \\ &\leq \xi_0^{1-p} - L'(1-p)s_1 - L'(1-p)(t-s_1) \\ &= \xi_0^{1-p} - L'(1-p)t. \end{aligned} \quad (11)$$

Continuing in this way, it can be obtained that on the interval $[s_{N-1}, s_N)$ (where N is given by eqn. (3))

$$\xi^{1-p}(t) \leq \xi_0^{1-p} - L'(1-p)t, \quad (12)$$

where $s_N < \mu T_p$. Further at $t = s_N$,

$$\begin{aligned} \xi^{1-p}(s_N) &\leq c^{1-p} \left\{ \xi_0^{1-p} - L'(1-p)s_N \right\} \\ &\leq \xi_0^{1-p} - L'(1-p)s_N. \end{aligned} \quad (13)$$

Now using eqn. (4) and (13), one can easily find that for all $t \in [s_N, s_{N+1})$

$$\begin{aligned} \xi^{1-p}(t) &\leq \xi^{1-p}(s_N) - L'(1-p)(t-s_N) \\ &\leq \xi_0^{1-p} - L'(1-p)s_N - L'(1-p)(t-s_N) \\ &= \xi_0^{1-p} - L'(1-p)t \\ &\leq \left(1 - (kl)^{\frac{1}{1-q}} \right)^{1-p} - L'(1-p)t \\ &= k' - L'(1-p)t, \end{aligned}$$

where $k' = \left(1 - (kl)^{\frac{1}{1-q}} \right)^{1-p}$. It should be noted that we have $s_N < \mu T_p < s_{N+1}$. Moreover, at the impulsive instant $t = s_{N+1}$ one can have

$$\xi^{1-p}(s_{N+1}) \leq c^{1-p} (k' - L'(1-p)s_{N+1}). \quad (14)$$

Also, for $t \in [s_{N+1}, s_{N+2})$ and using eqn. (14), we have

$$\begin{aligned} \xi^{1-p}(t) &\leq \xi^{1-p}(s_{N+1}) - L'(1-p)(t-s_{N+1}) \\ &\leq c^{1-p} (k' - L'(1-p)s_{N+1}) - L'(1-p)(t-s_{N+1}). \end{aligned}$$

Further, at the impulsive point $t = s_{N+2}$, one can have

$$\begin{aligned} \xi^{1-p}(s_{N+2}) &\leq c^{1-p} \left\{ c^{1-p} (k' - L'(1-p)s_{N+1}) - L'(1-p)(s_{N+2} - s_{N+1}) \right\} \\ &\leq c^{2-2p} k' - c^{1-p} L'(1-p)(s_{N+2} - s_{N+1}). \end{aligned} \quad (15)$$

Similarly, for $t \in [s_{N+2}, s_{N+3})$ it can be easily obtained that

$$\xi^{1-p}(t) \leq c^{2-2p} k' - c^{1-p} L'(1-p)(s_{N+2} - s_{N+1}) - L'(1-p)(t - s_{N+2}),$$

and at $t = s_{N+3}$

$$\xi^{1-p}(s_{N+3}) \leq c^{3-3p} k' - c^{2-2p} L'(1-p)(s_{N+2} - s_{N+1}) - c^{1-p} L'(1-p)(s_{N+3} - s_{N+2}).$$

Continuing in this way, it can be established that

$$\xi^{1-p}(s_{N+N_0}) \leq c^{N_0(1-p)} k' - c^{(N_0-1)(1-p)} L'(1-p)(s_{N+2} - s_{N+1}) - \dots - c^{1-p} L'(1-p)(s_{N+N_0} - s_{N+N_0-1}). \quad (16)$$

Now, using eqn. (5) in eqn. (16) it can be achieved that

$$\xi^{1-p}(s_{N+N_0}) \leq c^{N_0(1-p)} \{k' - (N_0 - 1) L'(1-p)\},$$

and using eqn. (6) it can be easily obtained that

$$\xi^{1-p}(s_{N+N_0}) \leq 0. \quad (17)$$

But, by eqn. (4) we have $s_{N+N_0} \leq T_p$. And, thus

$$\xi(T_p) \equiv 0, \forall t \geq T_p. \quad (18)$$

Case 2: $\xi_0 > 1 - (kl)^{\frac{1}{1-q}}$:

From conditions (iii) and (iv) one can see that

$$\begin{aligned} D^+ \xi(t) &\leq -A \xi(t)^P (1 - \xi(t)) + B \xi(t)^P (1 - \xi(t))^q \\ &\leq B \xi(t)^P (1 - \xi(t))^q. \end{aligned} \quad (19)$$

Further, when $\xi(t) \geq 1 - (kl)^{\frac{1}{1-q}}$ i.e., $\xi^{1-p}(t) \geq \left(1 - (kl)^{\frac{1}{1-q}}\right)^{1-p}$ and $(1 - \xi(t))^q \leq (kl)^{\frac{q}{1-q}}$, one can obtain from eqn. (19) that

$$\begin{aligned} D^+ \xi(t) &\leq B(kl)^{\frac{q}{1-q}} \xi(t) \xi^{p-1}(t) \\ &\leq \frac{B(kl)^{\frac{q}{1-q}}}{\left(1 - (kl)^{\frac{1}{1-q}}\right)^{1-p}} \xi(t) \\ &= B' \xi(t), \end{aligned} \quad (20)$$

where $B' = \frac{B(kl)^{\frac{q}{1-q}}}{\left(1 - (kl)^{\frac{1}{1-q}}\right)^{1-p}}$.

Now, we are going to construct the impulses in such a way that the trajectories of the system (1) reach the stable region provided in **Case 1** within time $\mu T_p < T_p$, where $\mu \in (0, 1)$.

For $t \in [0, s_1)$, one can have from eqn. (20) that

$$\begin{aligned} \xi(t) &\leq \xi_0 e^{B't} \\ &\leq e^{B't}, \end{aligned} \quad (21)$$

and at the impulsive point $t = s_1$, using the condition (ii), it can be obtained that

$$\xi(s_1) \leq c e^{B's_1}. \quad (22)$$

Similarly for $t \in [s_1, s_2)$, using eqn. (22), we have

$$\xi(t) \leq \xi(s_1) e^{B'(t-s_1)} = c e^{B't}.$$

And thus at the impulse $t = s_2$

$$\xi(s_2) \leq c^2 e^{B's_2}.$$

Proceeding similarly one can easily achieve that for $t \in [s_{N-1}, s_N)$

$$\xi(t) \leq \xi(s_{N-1}) e^{B'(t-s_{N-1})} = c^{N-1} e^{B't},$$

and thus, at $t = s_N$

$$\xi(s_N) \leq c^N e^{B's_N}. \quad (23)$$

Further, we have $s_N < \mu T_p$, and so whenever $t \in [s_N, \mu T_p]$ we can deduce that

$$\xi(t) \leq \xi(s_N) e^{B'(t-s_N)} = c^N e^{B't},$$

and thus

$$\xi(\mu T_p) \leq c^N e^{B'\mu T_p}. \quad (24)$$

Using eqn. (3) in the above eqn. (24), it can be easily deduced that

$$\xi(\mu T_p) \leq 1 - (kl)^{\frac{1}{1-q}}.$$

Then moving forward as in **Case 1**, we can easily have

$$\xi(T_p) \equiv 0, \forall t \geq T_p.$$

Thus, in both the cases we have $\xi(T_p) \equiv 0, \forall t \geq T_p$, which is equivalent to $V_P(y) \equiv 0, \forall t \geq T_p$. Moreover using condition (i) one can derive that

$$|y(t)| \leq \sigma_1^{-1}(V_P(y(t))) = 0, \forall t \geq T_p.$$

This completes the proof of Theorem 1. $\square$

Remark 2. It should be noted that, when we are considering the initial conditions y_0 from the stable region $\emptyset = \{y \in \mathbb{R}^n \mid \xi(t) \leq 1 - (kl)^{\frac{1}{1-q}}\}$, then the first N number of impulsive points, i.e., $s_1, s_2, \dots, s_N < \mu T_p$, does not play any role in establishing $U_G PT S$ of system (1) in Theorem 1. That is, if we only consider the impulses given in eqn. (4) satisfying provided sufficient conditions, we can achieve $U_G PT S$ within given prescribed-time T_p . In contrast, if our initial conditions y_0 are taken from outside the stable region $\emptyset$ then the first N number of impulsive points satisfying provided sufficient conditions, i.e., $s_1, s_2, \dots, s_N < \mu T_p$, are responsible to make the trajectories reach the region $\emptyset$ within the time $\mu T_p < T_p$. And after that the impulsive sequence given in eqn. (4) takes over to reach the origin within the prescribed-time T_p . Concisely we can say that, if we have the knowledge of the initial conditions and it is in the region $\emptyset$ then we can ignore the first N impulses $s_1, s_2, \dots, s_N < \mu T_p$ and still achieve our goal of $U_G PT S$ of the system (1) within the given prescribed-time T_p . But, more often than not initial conditions used to be unknown or poorly calculated. Thus, one may not be able to find either it belongs to the set $\emptyset$ or not. And hence we need total $N + N_0$ number of impulses satisfying given sufficient conditions.

Remark 3. From the right-hand side of the equations mentioned in conditions (iii) and (iv), one can notice that the vanishing term is contributed by the term ξ^p only, and $(1 - \xi)$ is not vanishing. Now, $f(0) = 0$ and ξ has the same power for both the conditions (iii) and (iv). So, $\varpi(t, y)$ is left with no choice but to be zero at $y = 0$ for all $t \geq 0$. Also, from Theorem 1, one can observe that we are able to reach the origin within some prescribed-time and stay there thereafter. This would have not been possible if $\varpi(t, 0) \neq 0, \forall t \geq 0$. This concludes that $\varpi(t, y)$ can only be vanishing disturbances here.

Remark 4. Let us estimate some forms of $\varpi(t, y)$ corresponding to the generally used Lyapunov form $V_P = |y|^h$, where $h \geq 1$. If the disturbance term $\varpi(t, y)$ satisfies $|\varpi(t, y)| \leq G V_P^{p - \left(\frac{h-1}{h}\right)} (\rho + \alpha V_P)^\delta$ for $t \geq 0$, where $G > 0, \rho \in (0, 1], \delta \in [0, 2 - p]$ and $p \in \left(\frac{h-1}{h}, 1\right)$, then it will satisfy condition (iv) of Theorem 1 for some adequately chosen values of the parameters. Moreover, the values of G, h, ρ, δ depend on the choices of the Lyapunov function, the form of $f(y)$ and the value of α present in $\xi(t)$. More insights on this regard are provided in Section 4.

Remark 5. A well known result on prescribed-time stability of nonlinear systems through impulsive control is given in [29]. However, this result was derived only for local region around the origin and also as can be observed from Remark 2 ([29]), the stable region was dependent on the given prescribed-time. Whereas, in the present manuscript we countered both the drawbacks, i.e., the prescribed-time stability result provided in Theorem 1 is global and the stable region provided here is independent of the given prescribed-time.

Theorem 2. For any prescribed-constant $T_p > 0$, let there exist κ_∞ functions σ_1, σ_2 , constants $\alpha, E, F, p_1, q_1, p_2, q_2 > 0$ with $p_1 > p_2, q_2 \geq q_1$ and a locally Lipschitz continuous function $V_P : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$, such that:

- i) $\sigma_1(|y|) \leq V_P(y) \leq \sigma_2(|y|)$;
 - ii) $\xi(V_P(g(y))) \leq c\xi(V_P(y))$ for $y \in \mathbb{R}^n$;
 - iii) $\frac{d\xi}{dV_P} \nabla V_P(y) \cdot f(y) \leq -E\xi^{p_1} (1 - \xi)^{q_1}$ for almost all $y \in \mathbb{R}^n$;
 - iv) $\frac{d\xi}{dV_P} \nabla V_P(y) \cdot \varpi(t, y) \leq F\xi^{p_2} (1 - \xi)^{q_2}$ for almost all $y \in \mathbb{R}^n$ and all $t \geq 0$;
- where, $\xi \equiv \xi(V_P) = \frac{\alpha V_P}{1 + \alpha V_P}$.

Then, system (1) is $U_G PPT S$ with respect to Ω and prescribed-time T_p over the class of impulsive sequences $\mathcal{H}_\Delta \subseteq \mathcal{H}$, where $\mathcal{H}_\Delta$ denotes the set of impulsive sequences $\{s_k\}$ with N_Δ as the minimum number of impulses satisfying

$$N_{\Delta} = \min \left\{ i \in \mathbb{N} \cup \{0\} \mid i \geq \frac{\ln \left((kl)^{\frac{1}{p_1-p_2}} \right) - Fk'T_p}{\ln(c)} \right\}, \text{ if } p_2 \geq 1; \quad (25)$$

or

$$N_{\Delta} = \min \left\{ i \in \mathbb{N} \cup \{0\} \mid i \geq \frac{\ln \left((kl)^{\frac{1}{p_1-p_2}} \right) - F'T_p}{\ln(c)} \right\}, \text{ if } p_2 < 1; \quad (26)$$

where $\Omega = \sigma_1^{-1} \left\{ \frac{1}{\alpha} \frac{(kl)^{\frac{1}{p_1-p_2}}}{1 - (kl)^{\frac{1}{p_1-p_2}}} \right\}$, $k' = \left\{ 1 - (kl)^{\frac{1}{p_1-p_2}} \right\}^{q_2}$, $F' = \frac{Fk'}{(kl)^{\frac{1-p_2}{p_1-p_2}}}$, $l = \frac{F}{E} < 1$ and $k > 1$ be such that $kl < 1$.

Proof. Let $T_p > 0$ be given prescribed-time. Further, for any initial data $y_0 \in \mathbb{R}^n$ and impulsive sequence $\{s_k\} \in N_{\diamond}$, let $y(t) = y(t, y_0)$ be the solution of the system (1) through $(0, y_0)$.

We will divide the proof into two cases as, $\xi(t) \leq (kl)^{\frac{1}{p_1-p_2}}$ and $\xi(t) > (kl)^{\frac{1}{p_1-p_2}}$, where, $\xi_0 \equiv \xi(V_P(y_0))$ and $\xi(t) \equiv \xi(V_P(y(t)))$.

Case 1: $\xi_0 \leq (kl)^{\frac{1}{p_1-p_2}}$:

Now, our claim is $\xi(t) \leq (kl)^{\frac{1}{p_1-p_2}}$ for all $t \geq 0$. If this is not true then there is some $t > 0$ such that $\xi(t) > (kl)^{\frac{1}{p_1-p_2}}$ holds. Let us denote $t^{\Delta} = \inf \left\{ t \geq 0 \mid \xi(t) > (kl)^{\frac{1}{p_1-p_2}} \right\}$. By the virtue of condition (ii) one can easily verify that t^{Δ} is not an impulse point. So, we have

that $\xi(t^{\Delta}) = (kl)^{\frac{1}{p_1-p_2}}$ and $\xi(t) \leq (kl)^{\frac{1}{p_1-p_2}}$ for all $t \in [0, t^{\Delta}]$ with $D^+\xi(t)|_{t=t^{\Delta}} \geq 0$, where $D^+\xi(t)$ is the derivative of ξ along the solution trajectories of the system (1). But, using the conditions (iii) and (iv) one can obtain

$$\begin{aligned} D^+\xi(t)|_{t=t^{\Delta}} &\leq -E\xi^{p_1}(t^{\Delta}) \left(1 - \xi(t^{\Delta})\right)^{q_1} + F\xi^{p_2}(t^{\Delta}) \left(1 - \xi(t^{\Delta})\right)^{q_2} \\ &= E\xi^{p_2}(t^{\Delta}) \left(1 - \xi(t^{\Delta})\right)^{q_1} \left\{ -\xi^{p_1-p_2}(t^{\Delta}) + \frac{F}{E} \left(1 - \xi(t^{\Delta})\right)^{q_2-q_1} \right\} \\ &\leq E\xi^{p_2}(t^{\Delta}) \left(1 - \xi(t^{\Delta})\right)^{q_1} \{-kl + l\} \\ &< 0, \end{aligned}$$

which is a contradiction. Hence, $\xi(t) \leq (kl)^{\frac{1}{p_1-p_2}}$ for all $t \geq 0$. Further, using the function ξ and condition (i) one can have $|y(t)| \leq \Omega$ for all $t \geq 0$.

Case 2: $\xi_0 > (kl)^{\frac{1}{p_1-p_2}}$:

By using the conditions (iii), (iv) and $\xi(t) \geq (kl)^{\frac{1}{p_1-p_2}}$ it can be obtained that

$$D^+\xi(t) \leq Fk'\xi^{p_2}(t), \forall t \geq 0; \quad (27)$$

where $k' = \left\{ 1 - (kl)^{\frac{1}{p_1-p_2}} \right\}^{q_2}$. We will now consider two sub-cases depending on $p_2 \geq 1$ or $p_2 < 1$.

Subcase 1: $p_2 \geq 1$. In this subcase eqn. (27) can be reduced to

$$D^+\xi(t) \leq Fk'\xi(t), \forall t \geq 0. \quad (28)$$

For $t \in [0, s_1)$ one can obtain from eqn. (28) that

$$\xi(t) \leq \xi_0 e^{Fk't} \leq e^{Fk't},$$

and using condition (ii), we have at $t = s_1$ that

$$\xi(s_1) \leq ce^{Fk's_1}.$$

Continuing in this way one can deduce that for $t \in [s_{N_{\Delta}}, T_p]$

$$\xi(t) \leq c^{N_{\Delta}} e^{Fk't}. \quad (29)$$

Now using the eqn. (25) in eqn. (29) one can achieve that at $t = T_p$

$$\xi(T_p) \leq (kl)^{\frac{1}{p_1-p_2}}. \quad (30)$$

Subcase 2: $p_2 < 1$. From eqn. (27) we can have

$$\begin{aligned}
D^+ \xi(t) &\leq F k' \xi(t) \xi^{p_2-1}(t) \\
&\leq \frac{F k'}{(kl)^{\frac{1-p_2}{p_1-p_2}}} \xi(t) \\
&= F' \xi(t),
\end{aligned}$$

where $F' = \frac{F k'}{(kl)^{\frac{1-p_2}{p_1-p_2}}}$. Then, proceeding similarly as in *Subcase 1* and by using the eqn. (26) one can easily obtain that

$$\xi(T_p) \leq (kl)^{\frac{1}{p_1-p_2}}. \quad (31)$$

So, from both the subcases we can see that the trajectories of the system (1) will reach the stable region $|y(t)| \leq \Omega$ for all $t \geq T_p$. It should be noted that, by properly adjusting the value of α one could modify the region Ω as per the requirement of the user.

Hence, combining both the cases one can conclude that the system (1) is $U_G PPTS$ with respect to Ω and prescribed-time T_p over the class of impulsive sequences $\mathcal{H}_\Delta \subseteq \mathcal{H}$ satisfying given sufficient conditions.

This completes the proof of the Theorem 2. $\square$

Remark 6. From conditions (iii), (iv) of Theorem 2 and using $\xi(t) \geq (kl)^{\frac{1}{p_1-p_2}}$ one can have:

$$\begin{aligned}
D^+ \xi(t) &\leq E \xi^{p_2}(t) (1 - \xi(t))^{q_1} \{-kl + l\} \\
&< 0.
\end{aligned}$$

That is, the system (1) is negative definite outside the stable region $\mathcal{G} = \left\{ y \in \mathbb{R}^n \mid \xi(t) \leq (kl)^{\frac{1}{p_1-p_2}} \right\}$, corresponding to Theorem 2.

Corollary 1. If the terms $F k' T_p$ and $F' T_p$ corresponding to eqns. (25) and (26) of Theorem 2 are ignored, then also Theorem 2 can be established where eqn. (25) and (26) will be replaced with

$$N_\Delta = \min \left\{ i \in \mathbb{N} \cup \{0\} \mid i \geq \frac{\ln \left((kl)^{\frac{1}{p_1-p_2}} \right)}{\ln(c)} \right\}. \quad (32)$$

Proof. From the definition of ξ function we have, $\xi(t) \leq 1$. So, for $t \in [0, s_1]$ one has $\xi(t) \leq 1$. Now using condition (ii), we have at $t = s_1$ that

$$\xi(s_1) \leq c. \quad (33)$$

By virtue of Remark 6 one can thus have

$$\xi(t) \leq c, \forall t \in [s_1, s_2].$$

Again, using condition (ii), we have at $t = s_2$ that

$$\xi(s_2) \leq c^2.$$

Continuing in this way one can obtain that $t \in [s_{N_\Delta}, T_p]$

$$\xi(t) \leq c^{N_\Delta}. \quad (34)$$

Combining eqn. (34) with (32) one can easily have $\xi(T_p) \leq (kl)^{\frac{1}{p_1-p_2}}$. In fact, $\xi(s_{N_\Delta}) \leq (kl)^{\frac{1}{p_1-p_2}}$. This completes the proof of the Corollary 1. $\square$

Remark 7. In view of Corollary 1 it can be observed that the number impulses required in this case is lesser than that of Theorem 2. Further, it should be noted that results like Corollary 1 can not be obtained for eqn. (3) of Theorem 1, as the trajectories of the system (1) outside the stable region $\emptyset = \left\{ y \in \mathbb{R}^n \mid \xi(t) \leq 1 - (kl)^{\frac{1}{1-q}} \right\}$ defined there may not be negative definite always.

Remark 8. Theorem 2 is mainly concerned with dealing with the non-vanishing disturbances (both bounded and unbounded), and so the concept of $U_G PPTS$ is used here. It should be noted that, as per the authors' knowledge, none of the articles present in the literature have considered the practical-finite/fixed/predefined or prescribed-time stability of impulsive systems considering *unbounded non-vanishing disturbances*. Specifically, let us estimate some forms of *non-vanishing disturbances* $\varpi(t, y)$ corresponding to the commonly used Lyapunov form $V_P = |y|^{h_1}$, where $h_1 > 1$.

- *For bounded non-vanishing disturbances:* Let $q_1 = 1$ and the disturbance term $\varpi(t, y)$ satisfy $|\varpi(t, y)| \leq \eta$ for some $\eta > 0$, then it satisfies conditions (iii) and (iv) of Theorem 2 corresponding to $p_2 = \frac{h_1-1}{h_1}$ and $q_2 = \frac{h_1+1}{h_1}$.
- *For unbounded non-vanishing disturbances:* Let $q_1 \leq 1$ and the disturbance term $\varpi(t, y)$ satisfy $|\varpi(t, y)| \leq H(\rho_1 + \alpha V_P)^{\delta_1}$, where $H > 0, 0 < \rho_1 \leq 1$ and $0 < \delta_1 \leq \left(\frac{1+h_1}{h_1}\right) - q_1$. As $f(0) = 0$ and $\varpi(t, y)$ is non-vanishing at the origin, $p_1 > p_2$ holds always. Further, the restriction $\delta_1 \leq \left(\frac{1+h_1}{h_1}\right) - q_1$ on δ_1 will provide $q_2 \geq q_1$. So, the *unbounded non-vanishing disturbances* of above form will satisfy the condition (iv) of Theorem 2 for some properly chosen parametric values.

Moreover, the values of the parameters used above will depend on the choices of Lyapunov function, the form of $f(y)$ and the value of α present in $\xi(t)$. More insight on this regard is provided in Section 4.

Remark 9. It should be noted that if one considers $h_1 = 1$ in Remark 8, then the disturbance can only be vanishing disturbances only, to satisfy the conditions of Theorem 2. Further, we would like to mention that the forms of the disturbances $\varpi(t, y)$ and the restrictions on the corresponding parameters used in Remarks 4 and 8 are some sufficient conditions that satisfy the provided Lyapunov-like conditions of Theorems 1 and 2. They are provided to give some insight to the efficacy of the proposed Lyapunov-like forms, and there can be several more estimations of the disturbance term $\varpi(t, y)$ other than those proposed in Remarks 4 and 8.

Remark 10. Article [36] has provided sufficient conditions for finite-time stability and practical finite-time stability of nonlinear systems consisting of vanishing and bounded disturbances via impulsive control. However, getting rid of the dependency on the initial conditions and constructing an impulsive sequence that can ensure the stability of the system (1) within a pre-specified time provided by the user is challenging work. Theorems 1 and 2 of the present work are dedicated to overcome this challenge.

Remark 11. In [38], some new results for the finite-time stability of impulsive systems have been studied, to speed up the convergence rate and reduce the settling-time. Though it has the upper hand, corresponding to some previous articles in this area (such as [17]), it also suffers from the drawback of dependency of the settling-time on the initial conditions. Moreover, it does not account for any kind of disturbances, which generally occur in every real-world problem. Further, [37] has proposed some Lyapunov-based sufficient conditions for practical finite-time stability of disturbed impulsive systems by virtue of delayed impulse control. However, similar to the above, it also has the disadvantage of reliance of the settling-time on the initial conditions. Also, it accounts only for the bounded disturbances. In contrast, in the present article, not only is the dependency of the settling-time on the initial conditions removed but it can also be pre-specified by the users according to the requirements. Moreover, the present manuscript addresses not only the bounded disturbances but also some specific forms of the unbounded vanishing and unbounded non-vanishing disturbances via Theorems 1 and 2.

Remark 12. The authors of the article [36] have considered the Lyapunov function to be globally Lipschitz continuous, which is a very strict condition. Thus left with the only choice of $|x|$ as the Lyapunov function. In this manuscript, no such restrictions on Lyapunov function is provided, and thereby, the range of selection of Lyapunov functions is much broader here. Hence, it is believed that the results provided here will be beneficial for several studies on this aspect.

Remark 13. It should be noted that though the continuous unperturbed and discrete dynamics considered here are stable, continuous dynamics is of indefinite form when we consider the perturbation term into the account. Hence, impulses acting here not only reduce the settling time, as per the requirements, but also act as controllers to make the unstable system stable. Elaborately, in **Case 1** of Theorem 1 and in Theorem 2, the impulses are designed to reduce the settling time according to the requirements of the user, whereas, in **Case 2** of Theorem 1, the impulses are acting as controllers. Further, in place of stable discrete dynamics, if we had considered hybrid discrete dynamics then it would have been more difficult to obtain U_GPTS and U_GPPTS . This will be our future goal.

Remark 14. In contrast to the several recent articles on practical finite-time and finite-time stability [13,33,34,39], which are based on continuous dynamics of the system, in the present study, we have established U_GPTS and U_GPPTS of disturbed impulsive systems corresponding to three different kinds of disturbances (vanishing disturbances, bounded non-vanishing disturbances and unbounded non-vanishing disturbances). Here the main focus has not only been on the stability of hybrid systems (continuous and discrete) but also to enhance the stability from finite/practical-finite time to prescribed/practical prescribed-time. Moreover, the construction of Lyapunov-like sufficient conditions along with the formation of impulsive sequences to make the impulsive system stable within the prescribed-time were the main challenges of the work.

4. Numerical simulations

In this section, three examples are provided to validate the truthfulness of the results of the present study obtained in the previous sections.

Example 1. Let us consider a ball-motion model with external perturbations and sudden changes in the velocity as follows:

$$\begin{cases} \dot{y}(t) = -\frac{A}{2} \operatorname{sgn}(y(t)) |y(t)|^{\frac{1}{3}} (1 + y^2(t))^{\frac{1}{3}} + \varpi(t, y(t)), & t \neq s_k, \\ y(t) = \left\{ \frac{cy^2(t^-)}{1+(1-c)y^2(t^-)} \right\}^{\frac{1}{2}}, & t = s_k, \end{cases} \quad (35)$$

where $y(t)$ denotes the velocity of the ball and $\varpi(t, y(t)) = \frac{1}{4} \sin(t) |y(t)|^{\frac{1}{3}} (1 + y^2(t))^{\frac{13}{12}}$ represents the external perturbation due to the friction. Let us consider the Lyapunov function as $V_P(y) = y^2$ and $\xi(V_P) = \frac{\alpha V_P}{1+\alpha V_P}$ with $\alpha = 1$. Then the system (35) satisfies the conditions (i)-(iv) of Theorem 1 with $\sigma_1(|y|) = \sigma_2(|y|) = y^2$, $A = 1$, $B = \frac{1}{2}$, $l = \frac{1}{2}$, $p = \frac{2}{3}$, $q = \frac{1}{4}$. Also, as per Remark 4, we have

$G = 1$, $h = \frac{1}{3}$, $\rho = 1$ and $\delta = \frac{13}{12}$. Hence, one can easily obtain the stable region as $|y(t)| \leq \sqrt{\left(\frac{4}{3}\right)^{\frac{4}{3}} - 1} \approx 0.68$. Further, using eqns. (3)

and (6), $\mu = \frac{1}{2}$ and $k = \frac{3}{2}$, one can have $N \geq 1$ and $N_0 \geq 11$. For the purpose of numerical simulation, let us consider the Prescribed-time as $T_p = 2$, impulsive strength $c = \frac{1}{20}$, $N = 1$, $N_0 = 11$. Now, we choose the initial condition as $y_0 = 2$ and the impulsive sequence $\mathcal{H}_\diamond = \{s_k\}$, satisfying eqns. (2), (4) and (5), as $\{0.5, 1.01, 1.38, 1.52, 1.57, 1.589, 1.596, 1.6, 1.61, 1.62, 1.63, 1.7\}$. Fig. 1 justifies the result provided in Theorem 1. Meanwhile, if we choose the initial condition within the derived stable region (for example $y_0 = 0.5 \leq 0.68$) then we only required $N_0 = 11$ number of impulse, i.e., we can leave the first N number of impulses, satisfying eqns. (4) and (5) (for example $\{1.01, 1.38, 1.52, 1.57, 1.589, 1.596, 1.6, 1.61, 1.62, 1.63, 1.7\}$) to reach the origin within the provided prescribed-time $T_p = 2$. Fig. 1b validates it and this verifies the statement of Remark 2.

Example 2. In this example we are going to consider one-dimensional impulsive system with non-vanishing bounded perturbations as:

$$\begin{cases} \dot{y}_1(t) = -\frac{E}{2} |y_1(t)|^{\frac{1}{2}} \operatorname{sgn}(y_1(t)) + \varpi(t, y_1(t)), & t \neq s_k, \\ y_1(t) = \left\{ \frac{cy_1^2(t^-)}{1+(1-c)y_1^2(t^-)} \right\}^{\frac{1}{2}}, & t = s_k, \end{cases} \quad (36)$$

where E is a positive real number and $\varpi(t, y_1(t)) = 0.25 \cos(t)$ is the external disturbances which is non-vanishing and bounded. Considering the Lyapunov function as $V_P(y_1) = y_1^2$ and $\xi(V_P) = \frac{\alpha V_P}{1+\alpha V_P}$ with $\alpha = 1$ one can obtain that the system (36) satisfies the conditions (i)-(iv) of Theorem 2 with $E = 1$, $F = \frac{1}{2}$, $p_1 = \frac{3}{4}$, $q_1 = \frac{5}{4}$, $p_2 = \frac{1}{2}$ and $q_2 = \frac{3}{2}$. Also, we have $l = \frac{F}{E} = \frac{1}{2} < 1$, $p_1 > p_2$ and $q_1 \leq q_2$.

Further, considering the impulsive strength $c = \frac{1}{2}$ and $k = \frac{3}{2}$ one can find out that the stable region is given by $|y_1(t)| \leq \sqrt{\frac{\left(\frac{3}{4}\right)^4}{1-\left(\frac{3}{4}\right)^4}} \approx 0.68$. Also, N_Δ can be calculated as $N_\Delta \geq 4$. For, simulation purpose, we will be considering the prescribed-time constant $T_p = 2$, number of impulses $N_\Delta = 4$ and the corresponding impulsive sequence is considered as $\{1, 1.2, 1.3, 1.5\}$. One can see from Fig. 2 that the trajectory without impulses is not able to reach the provided stable region within the prescribed-time $T_p = 2$, whereas, it is possible corresponding to the provided number of impulses. Hence, Fig. 2 validates the claims of the Theorem 2 corresponding to non-vanishing bounded perturbations.

Example 3. Let us consider a one-dimensional impulsive system with non-vanishing unbounded disturbances as:

$$\begin{cases} \dot{y}_2(t) = -\frac{E}{2} |y_2(t)|^{\frac{2}{3}} \operatorname{sgn}(y_2(t)) (1 + y_2^2(t)) + \varpi(t, y_2(t)), & t \neq s_k \\ y_2(t) = \left\{ \frac{cy_2^2(t^-)}{1+(1-c)y_2^2(t^-)} \right\}^{\frac{1}{2}}, & t = s_k, \end{cases} \quad (37)$$

where, E is a positive number and the non-vanishing unbounded disturbance is considered as $\varpi(t, y_2(t)) = 0.25 \sin(t) (1 + y_2^2)^{\frac{4}{3}}$.

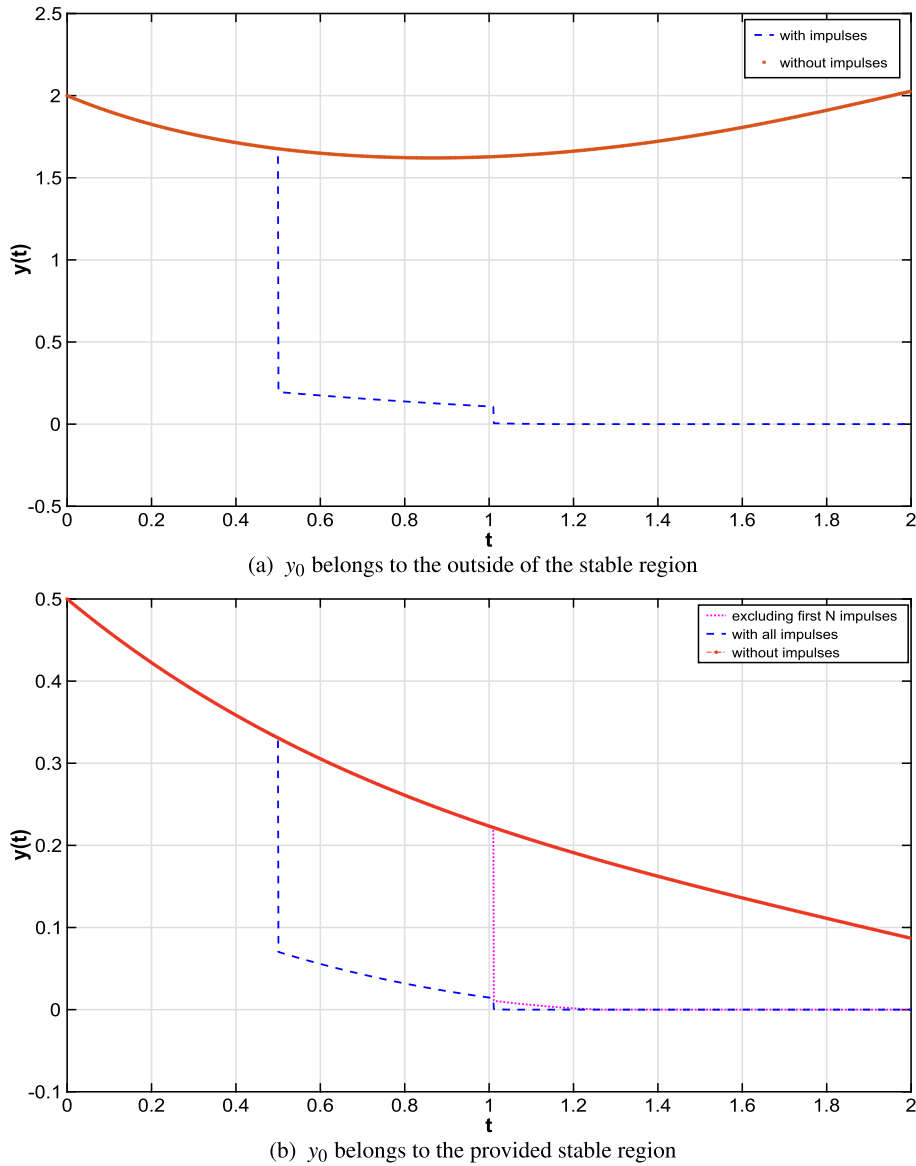


Fig. 1. Simulations of Example 1.

Then, it satisfies the conditions of Remark 8 with $H = 1, \rho_1 = 1, \delta_1 = \frac{4}{3}$. Now, consider the Lyapunov function $V_P(y_2) = y_2^2$ and $\xi(V_P) = \frac{\alpha V_P}{1 + \alpha V_P}$ with $\alpha = 1$. Then, the system (37) satisfies the conditions (i)-(iv) of Theorem 2 with $E = 1, F = \frac{1}{2}, p_1 = \frac{5}{6}, q_1 = q_2 = \frac{1}{6}, p_2 = \frac{1}{2}$. Clearly we have, $l = \frac{F}{E} = \frac{1}{2} < 1, p_1 > p_2$ and $q_1 \leq q_2$. Further, considering $k = \frac{3}{2}$, impulsive strength as $c = \frac{1}{3}$ and the prescribed-time constant $T_p = 1$ one can easily find $N_\Delta \geq 2$. Also, we have the stable region as $|y_2(t)| \leq \sqrt{\frac{27}{37}} \simeq 0.85$. For numerical purposes let $N_\Delta = 2$ with the corresponding impulsive sequence as $\{0.5, 0.6\}$. It can be seen from Fig. 3 that, the trajectory of the system (37) is not reaching the established stable region in the provided prescribed-time $T_p = 1$. Whereas, once we consider the impulses into account, it fulfils our objective. Hence, Fig. 3 authenticates the result of Theorem 2 corresponding to *non-vanishing unbounded disturbances*.

Remark 15. Consider the Example 3 keeping all the parameters same. Then by Corollary 1 we have $N_\Delta \geq 1$. For numerical purpose, let us consider $N_\Delta = 1$. Consider different sets impulsive sequence as $\{0.1\}, \{0.3\}, \{0.5\}, \{0.7\}, \{0.9\}$. Then from Fig. 4 one can see that for each case we have reached the stable region $|y_2(t)| \leq \sqrt{\frac{27}{37}} \simeq 0.85$ within the prescribed-time $T_p = 1$ and stayed there

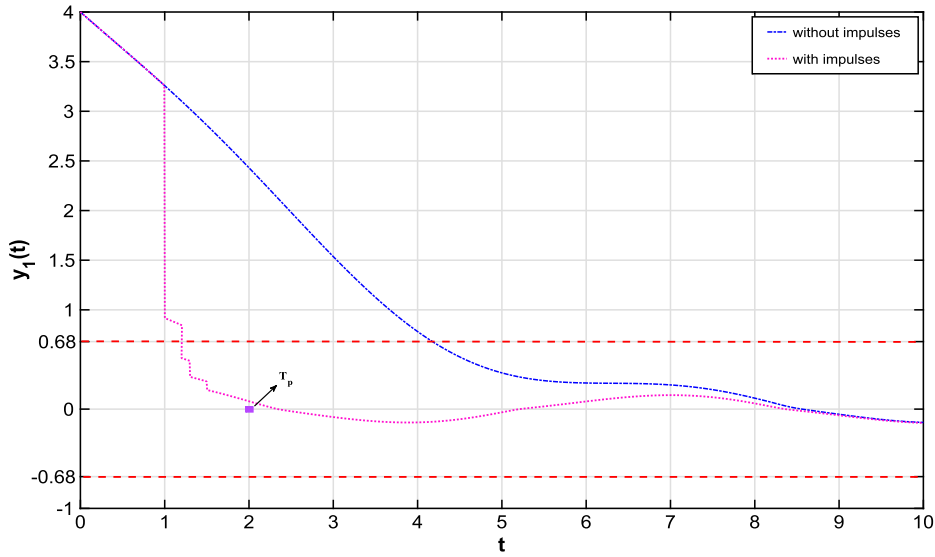


Fig. 2. Simulations of Example 2.

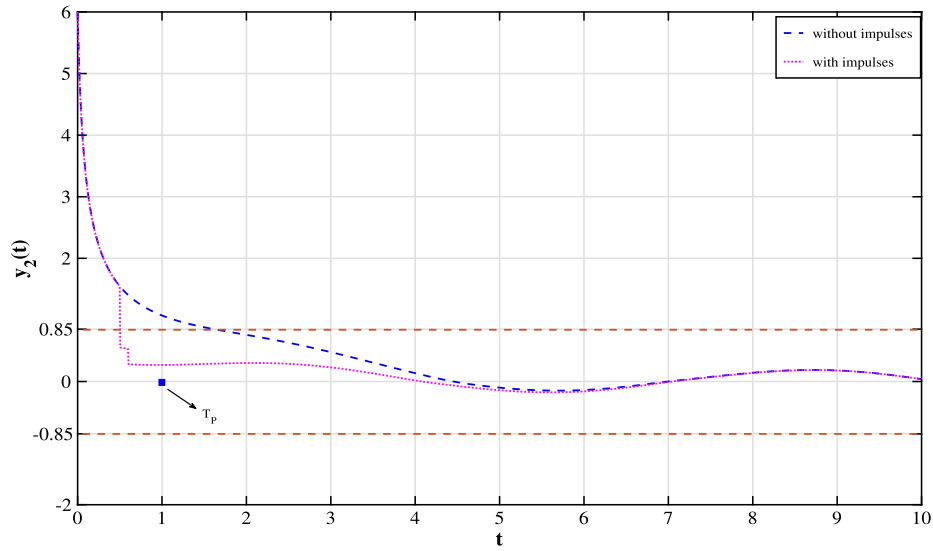


Fig. 3. Simulations of Example 3.

afterwards. This validates the Corollary 1. It can further be observed that the instances of occurrence of impulses, i.e., either they are nearer or away from the given prescribed time, do not affect the desired results.

5. Conclusion

Prescribed-time control of perturbed non-linear systems, consisting of vanishing and non-vanishing perturbations, is the main content of the article. Some Lyapunov-like sufficient conditions are derived for $U_G PTS$ and $U_G PPTS$ of perturbed impulsive systems. In contrast to the existing articles on impulsive systems, which can only handle bounded disturbances, this manuscript can address the effect of **non-vanishing unbounded disturbances**. Impulsive sequences constructed here consider two major aspects: one is to make the trajectories outside the provided stable region reach the stable region, and the other is to regulate the settling-time according to the users' choice. Further, the essentiality of the global Lipschitz condition of the Lyapunov function, as mentioned in the article [36], is weakened here. In the future, we will focus on providing such results for delayed impulsive systems as well as for impulsive systems with delayed impulses.

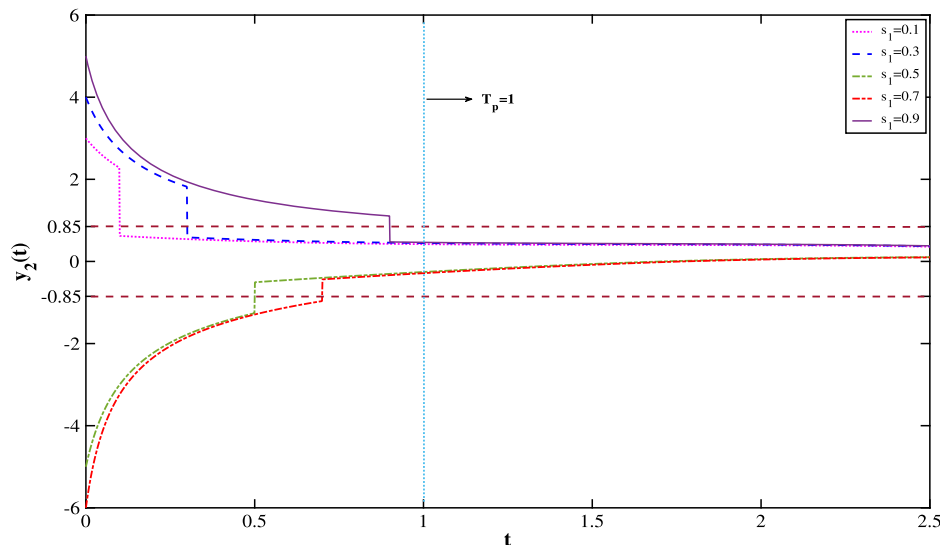


Fig. 4. Simulations of Remark 15 with respect to initial conditions 3, 4, -5, -6, 5.

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Data availability

No data was used for the research described in the article.

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